

DATABASES – COURSE 4 Database Protection & Relational DB

1. DB protection

DB protection = a set of measures necessary to ensure data security and integrity.

Data security = denial of access to data to unauthorized users.

Data integrity = correctness of loaded and manipulated data, that follows the data integrity constraints of the implemented data model.

INTEGRITY

1. *Semantic integrity* = prevents the inserting of incorrect data in the database and the achievement of incorrect processing, by following the constraints of integrity. They can be implicit (automatically insured by the DBMS) and explicit (provided by procedures included in application programs).

2. *Concurrency control* = ensuring data coherence and simultaneity for transaction processing (logic processing unit) by specific techniques (blocking, interlocking).

Ex. SELECT FOR UPDATE with WAIT clause, LOCK tables or accounts.

3. *Backup and recovery*

Backup = storing data backups through specific techniques (copying, logging). Backup -> archiving -> virtualization

Recovery = DB's restore consistency, starting from saved data by minimizing the loss of processed data (automatic / manual).

SECURITY

1. *Data authorization and access control* = identifies the users and restricts access to different processing operations.

2. *Views* = logical database partitions defined for different users. CREATE VIEW...AS SELECT...[WITH READ ONLY]

3. *Special procedures* = runtimes provided by the DBMS, which perform certain operations on data and that are accessible to some users.

4. *Encryption* = operations for data coding that are used for storage or transmission of data (through passwords, encrypting/decrypting algorithms).

2. Relational database (RDB)

- The concept of RDB
- RDB Design using normalization technique
- Relational Algebra and Calculus

The concept of RDB

RDB use the relational data model and the related concepts.

RDB have a solid theoretical foundation, especially through IBM researches that have been conducted by E.F.CODD.

RDB = an organized set of tables (relations) with links between them.

Some advantages for using RDB instead of files:

- provide logical and physical data independence;
- structure data on three levels: conceptual, logical and physical;
- simplify the representation and usage of data (through the data model);

- ensure openness and portability of applications;
- data structure is stored in the database dictionary, not in programs.

Comparative concepts regarding the organization of data in files and in RDB:

Files	→ File	→ Record	→ Field	→ Values
RDB	→ Table (relation)	→ Tuple (line)	→ Attribute (column)	→ Domain of values

Minimal conditions for a DBMS to be relational:

- to implement the relational data model through DDL and DML;
- to implement at least one relational language through DML.

The complete terms for a relational DBMS are given by Codd's 13 rules.

RDB Design using normalization technique

a) About normalization

The dynamic aspect of the data structures is solved by RDB – DB schema is being modified without generating anomalies.

The normalization technique is used in the design of the RDB structure and helps to eliminate the structure update anomalies (deficiencies).

Update anomalies are undesirable situations that may arise in certain tables in their designing process:

- delete anomalies = by deleting a tuple in a table, also relevant information may be lost, in addition to deleting the desired information;
- adding anomalies = new information cannot be included in a table, because one does not know other useful information (for example, the key values or any other value that needs to be not null); EX: STUDENT_NAME, STUDENT_ID, STUDENT_GROUP, TEACHER_NAME, TEACHER_ADDRESS
- modify anomaly = is difficult to change a value of an attribute when it appears in several tuples.

Update anomalies are resolved by relational theory through design techniques, which does not happen to other types of DB.

Normalization = the theory built around the concept of normal forms (NF), which improves the DB structure by gradually removing the deficiencies and by providing improved facilities for manipulation of data.

The normalization theory was originally conceived by E.F. CODD and later developed by other researchers.

The normalization is using the top-down method by splitting a table into two or more tables, keeping the link information (attributes).

b) Normal Forms

A table (relation) is in a normal form (NF) if it meets certain constraints, indicating that the NF($i + 1$) is preferred to the NF(i), $i = 1, 4$.

There are five normal forms (NF1 to NF5) and one additional (BCNF), revising NF3.

A database is in NF i , $i = 1, 5$ if all its tables are in NF i .

The initial DB will be formed of a single data collection that will afterwards split in a number of tables (through projection), following the normal forms.